

Partial Progress on Coderivative-Free Lipschitz-Modulus Bounds for Regularized Least Squares

Prepared mathematical note

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Abstract

This note gives a corrected and deliberately limited response to the quantitative question raised by Cui, Hoheisel, Nghia, and Sun concerning the Lipschitz modulus of the solution mapping for convex regularized least-squares problems. The status is explicit: this is *not* a solution of the exact modulus-computation problem. The positive result is a local upper bound obtained from a secant monotonicity modulus of the first-order graph $\text{gph } \partial g$. The proof is a direct strong-monotonicity comparison and does not use coderivatives. Two limitations are then proved. First, in smooth problems the estimate reduces to the generally nonsharp bound $\|B\|/\lambda_{\min}(H)$, whereas the exact modulus is $\|H^{-1}B\|$; the gap can be arbitrarily large. Second, the exact modulus cannot be determined from only the pointwise first-order data $\partial g(\bar{x})$, $\partial g^*(\bar{u})$, and $\text{par } \partial g^*(\bar{u})$. Thus the results are partial progress toward conservative certificates and toward identifying insufficient information, not a full answer to the authors' open problem.

1 Scope and setting

Cui, Hoheisel, Nghia, and Sun studied Lipschitz stability of the solution mapping

$$S(A, b, \mu) := \operatorname{argmin}_{x \in \mathbb{X}} \left\{ \frac{1}{2\mu} \|Ax - b\|^2 + g(x) \right\} \quad (1)$$

under assumptions ensuring that g^* is C^2 -cone reducible; they left open the quantitative problem of computing the Lipschitz modulus without difficult second-order variational information about g [1]. The present note does not solve that problem. It gives an upper estimate and proves that this estimate is not exact in general.

Throughout, $\mathbb{X}$ and $\mathbb{Y}$ are finite-dimensional Euclidean spaces, $g : \mathbb{X} \rightarrow \mathbb{R} \cup \{+\infty\}$ is closed, proper, and convex, and

$$P := \mathcal{L}(\mathbb{X}, \mathbb{Y}) \times \mathbb{Y} \times (0, \infty), \quad p = (A, b, \mu).$$

The parameter space is equipped with

$$\|(\Delta A, \Delta b, \Delta \mu)\|^2 := \|\Delta A\|_F^2 + \|\Delta b\|^2 + |\Delta \mu|^2.$$

Define

$$Q(p, x) := \frac{1}{\mu} A^*(Ax - b). \quad (2)$$

Then $x \in S(p)$ if and only if

$$0 \in Q(p, x) + \partial g(x). \quad (3)$$

Fix a reference triple $\bar{p} = (\bar{A}, \bar{b}, \bar{\mu})$ and a reference solution $\bar{x} \in S(\bar{p})$, and put

$$\bar{u} := -Q(\bar{p}, \bar{x}) \in \partial g(\bar{x}). \quad (4)$$

If s is a single-valued localization of S around $(\bar{p}, \bar{x})$, its local Lipschitz modulus is

$$\text{lip } s(\bar{p}) := \limsup_{\substack{p, q \rightarrow \bar{p} \\ p \neq q}} \frac{\|s(p) - s(q)\|}{\|p - q\|}.$$

2 A local secant modulus

For $\rho > 0$, define

$$\mathbf{m}_\rho := \inf \left\{ \frac{\mu^{-1} \|A(x - y)\|^2 + \langle u - v, x - y \rangle}{\|x - y\|^2} : \begin{aligned} & \|p - \bar{p}\| \leq \rho, \quad p = (A, b, \mu), \quad (x, u), (y, v) \in \text{gph } \partial g, \quad x \neq y, \\ & \|x - \bar{x}\| + \|y - \bar{x}\| + \|u - \bar{u}\| + \|v - \bar{u}\| \leq \rho \end{aligned} \right\}. \quad (5)$$

Set

$$\mathbf{m}_* := \liminf_{\rho \downarrow 0} \mathbf{m}_\rho. \quad (6)$$

Also define

$$K_\rho := \sup \left\{ \frac{\|Q(p, x) - Q(q, x)\|}{\|p - q\|} : \|p - \bar{p}\|, \|q - \bar{p}\| \leq \rho, \|x - \bar{x}\| \leq \rho, p \neq q \right\}. \quad (7)$$

Since Q is C^1 in (p, x) for $\mu > 0$,

$$K_\rho \rightarrow K_* := \|D_p Q(\bar{p}, \bar{x})\| \quad (\rho \downarrow 0). \quad (8)$$

Writing $\bar{r} := \bar{A}\bar{x} - \bar{b}$, the derivative $B := D_p Q(\bar{p}, \bar{x})$ is

$$B(\Delta A, \Delta b, \Delta \mu) = \frac{1}{\bar{\mu}} \left(\Delta A^* \bar{r} + \bar{A}^* (\Delta A \bar{x} - \Delta b) \right) - \frac{\Delta \mu}{\bar{\mu}^2} \bar{A}^* \bar{r}. \quad (9)$$

Remark 2.1. *The quotient in (5) is nonnegative by monotonicity of ∂g . Positivity of $\mathbf{m}_\rho$ is a local uniform strong-monotonicity condition for $x \mapsto Q(p, x) + \partial g(x)$. It is a sufficient quantitative condition, not a necessary characterization of Lipschitz stability.*

3 The upper estimate

The next theorem is the main valid estimate. It is stated for solutions already known to exist in the chosen localization; local existence of a branch must come from a separate stability or existence argument.

Theorem 3.1 (Localized secant estimate). *Fix $\rho > 0$ with $\mathbf{m}_\rho > 0$. Let $p = (A, b, \mu)$ and $q = (\tilde{A}, \tilde{b}, \tilde{\mu})$ satisfy $\|p - \bar{p}\|, \|q - \bar{p}\| \leq \rho$. Let $x \in S(p)$ and $y \in S(q)$, and define*

$$u := -Q(p, x) \in \partial g(x), \quad v := -Q(q, y) \in \partial g(y).$$

Assume that

$$\|x - \bar{x}\| + \|y - \bar{x}\| + \|u - \bar{u}\| + \|v - \bar{u}\| \leq \rho. \quad (10)$$

Then

$$\|x - y\| \leq \frac{K_\rho}{\mathfrak{m}_\rho} \|p - q\|. \quad (11)$$

In particular, for fixed p there is at most one solution satisfying the localization condition.

Proof. Put $h := x - y$. If $h = 0$, there is nothing to prove. Suppose $h \neq 0$. From the definition of $\mathfrak{m}_\rho$ and (10),

$$\mathfrak{m}_\rho \|h\|^2 \leq \frac{1}{\mu} \|Ah\|^2 + \langle u - v, h \rangle. \quad (12)$$

Because $Q(p, x) - Q(p, y) = \mu^{-1} A^* A(x - y)$, and because $u = -Q(p, x)$ and $v = -Q(q, y)$,

$$\begin{aligned} \frac{1}{\mu} \|Ah\|^2 + \langle u - v, h \rangle &= \langle Q(p, x) - Q(p, y) + u - v, h \rangle \\ &= \langle Q(q, y) - Q(p, y), h \rangle. \end{aligned} \quad (13)$$

Combining (12) and (13) gives

$$\mathfrak{m}_\rho \|h\|^2 \leq \|Q(q, y) - Q(p, y)\| \|h\| \leq K_\rho \|p - q\| \|h\|.$$

Dividing by $\mathfrak{m}_\rho \|h\|$ proves (11). If $p = q$, the estimate gives $\|x - y\| = 0$, proving localized uniqueness. $\square$

Corollary 3.2 (Bound for an existing branch). *Assume $\mathfrak{m}_* > 0$. Suppose that S has a single-valued localized branch s around $(\bar{p}, \bar{x})$ and that the corresponding dual element $-Q(p, s(p))$ remains near $\bar{u}$ in this localization. Then*

$$\text{lip } s(\bar{p}) \leq \frac{\|D_p Q(\bar{p}, \bar{x})\|}{\mathfrak{m}_*} = \frac{\|B\|}{\mathfrak{m}_*}. \quad (14)$$

Proof. For every $\varepsilon > 0$, the definition of $\liminf$ gives $\mathfrak{m}_\rho \geq \mathfrak{m}_* - \varepsilon$ for all sufficiently small $\rho > 0$. Also $K_\rho \rightarrow \|B\|$ by (8). Apply Theorem 3.1 to $x = s(p)$ and $y = s(q)$ for p, q close enough to $\bar{p}$, then let $p, q \rightarrow \bar{p}$, $\rho \downarrow 0$, and $\varepsilon \downarrow 0$. $\square$

Remark 3.3. *Corollary 3.2 is not a complete stability theorem. It does not prove nonemptiness of $S(p)$ near $\bar{p}$. It gives a Lipschitz estimate once a localized solution branch is known to exist, for example from a separate strong-regularity theorem or from direct model-specific arguments.*

4 Smooth case: the exact modulus and the nonsharp bound

The secant estimate becomes a familiar but generally nonsharp bound in smooth problems.

Proposition 4.1 (Smooth secant value). *Assume g is C^2 near $\bar{x}$ and define*

$$H := \frac{1}{\bar{\mu}} \bar{A}^* \bar{A} + \nabla^2 g(\bar{x}). \quad (15)$$

Then

$$\mathfrak{m}_* = \lambda_{\min}(H). \quad (16)$$

Consequently, if H is positive definite, Corollary 3.2 gives the upper bound

$$\text{lip } s(\bar{p}) \leq \frac{\|B\|}{\lambda_{\min}(H)}. \quad (17)$$

Proof. Near $(\bar{x}, \bar{u})$, $u = \nabla g(x)$ and $v = \nabla g(y)$. For $h = x - y \neq 0$ and $e = h/\|h\|$, the quotient in (5) is

$$\langle \mu^{-1} A^* A e, e \rangle + \int_0^1 \langle \nabla^2 g(y + th) e, e \rangle dt.$$

As $\rho \downarrow 0$, the matrices in this display converge uniformly to $\bar{\mu}^{-1} \bar{A}^* \bar{A}$ and $\nabla^2 g(\bar{x})$. Taking the infimum over unit vectors e yields (16). $\square$

Proposition 4.2 (Exact smooth modulus). *Assume g is C^2 near $\bar{x}$ and that H in (15) is nonsingular. Then the localized solution branch is C^1 near $\bar{p}$, and its exact local Lipschitz modulus is*

$$\text{lip } s(\bar{p}) = \|H^{-1} B\|. \quad (18)$$

Proof. Let $F(p, x) := Q(p, x) + \nabla g(x)$. Then $F(\bar{p}, \bar{x}) = 0$, $D_x F(\bar{p}, \bar{x}) = H$, and $D_p F(\bar{p}, \bar{x}) = B$. The classical implicit function theorem gives a C^1 branch s with

$$Ds(\bar{p}) = -H^{-1} B.$$

For a C^1 map between finite-dimensional normed spaces, the local Lipschitz modulus at a point equals the operator norm of the derivative at that point. Therefore $\text{lip } s(\bar{p}) = \|H^{-1} B\|$. $\square$

The bound (17) is merely the inequality

$$\|H^{-1} B\| \leq \|H^{-1}\| \|B\| = \frac{\|B\|}{\lambda_{\min}(H)}$$

when H is positive definite. Equality need not hold.

Theorem 4.3 (The gap can be arbitrarily large). *For every $0 < \varepsilon < 1$, there is a smooth strongly convex quadratic example for which the exact local Lipschitz modulus is $1/2$, while the secant bound (14) equals $1/\varepsilon$. Hence the ratio between the bound and the exact modulus is $2/\varepsilon$.*

Proof. Let $\mathbb{X} = \mathbb{R}^2$, $\mathbb{Y} = \mathbb{R}$, and

$$\bar{A} = (1, 0), \quad \bar{b} = 0, \quad \bar{\mu} = 1.$$

For $0 < \varepsilon < 1$, set

$$g_\varepsilon(x_1, x_2) = \frac{1}{2} x_1^2 + \frac{\varepsilon}{2} x_2^2.$$

Then $\bar{x} = 0$, $\bar{u} = 0$, and

$$H = \bar{A}^* \bar{A} + \nabla^2 g_\varepsilon(0) = \begin{pmatrix} 2 & 0 \\ 0 & \varepsilon \end{pmatrix}.$$

Because $\bar{r} = \bar{A}\bar{x} - \bar{b} = 0$ and $\bar{x} = 0$, formula (9) gives

$$B(\Delta A, \Delta b, \Delta \mu) = -\bar{A}^* \Delta b = (-\Delta b, 0).$$

Thus $\|B\| = 1$ and, by Proposition 4.1, $\mathbf{m}_* = \lambda_{\min}(H) = \varepsilon$. The secant bound is therefore $1/\varepsilon$.

On the other hand, Proposition 4.2 gives

$$H^{-1} B(\Delta A, \Delta b, \Delta \mu) = (-\Delta b/2, 0),$$

so $\|H^{-1} B\| = 1/2$ with respect to the product parameter norm. The claimed ratio follows. $\square$

5 A no-go result for pointwise first-order data

The next theorem shows that the exact modulus cannot be recovered from only pointwise first-order data at the reference point.

Theorem 5.1 (Pointwise first-order data are insufficient). *There is no function of only*

$$\bar{p}, \quad \bar{x}, \quad \bar{u}, \quad \partial g(\bar{x}), \quad \partial g^*(\bar{u}), \quad \text{par } \partial g^*(\bar{u}) \quad (19)$$

that equals the exact local Lipschitz modulus of S for all closed proper convex regularizers for which the localized solution map is single-valued and Lipschitz.

Proof. Take $\mathbb{X} = \mathbb{Y} = \mathbb{R}$ and

$$\bar{A} = 1, \quad \bar{b} = 0, \quad \bar{\mu} = 1, \quad \bar{x} = 0.$$

For each $a > 0$, define

$$g_a(x) = \frac{a}{2}x^2, \quad g_a^*(u) = \frac{1}{2a}u^2.$$

Then $\bar{u} = 0$ and, for every $a > 0$,

$$\partial g_a(0) = \{0\}, \quad \partial g_a^*(0) = \{0\}, \quad \text{par } \partial g_a^*(0) = \{0\}.$$

Thus all the data in (19) are identical throughout the family.

But the exact modulus depends on a . With $A = 1$ and $\mu = 1$ fixed,

$$S_a(1, b, 1) = \operatorname{argmin}_x \left\{ \frac{1}{2}(x - b)^2 + \frac{a}{2}x^2 \right\} = \frac{b}{1 + a}.$$

Hence the local Lipschitz modulus with respect to b is $1/(1 + a)$. The same value is the full parameter modulus at the reference point, because $D_p Q(\bar{p}, 0)$ depends only on Δb when $\bar{x} = \bar{b} = 0$. Since a can vary over $(0, \infty)$, identical pointwise data produce different exact moduli. No function of only those data can compute the exact modulus in general. $\square$

Remark 5.2. *Theorem 5.1 is narrow. It rules out exact formulas based only on the pointwise sets in (19). It does not rule out exact formulas based on richer local information, such as the local graph of ∂g . In the example above, $\text{gph } \partial g_a = \{(x, ax) : x \in \mathbb{R}\}$ immediately reveals a .*

6 Conclusion

This note obtains partial progress only. The valid positive result is the certified upper bound

$$\text{lip } s(\bar{p}) \leq \frac{\|D_p Q(\bar{p}, \bar{x})\|}{\mathbf{m}_*}$$

for an already-existing localized branch. The bound is coderivative-free in notation and can be interpreted as a local graph-secant monotonicity certificate.

The limitations are essential. The estimate is not an exact Lipschitz-modulus formula; in smooth problems the exact modulus is $\|H^{-1}B\|$, while the secant estimate gives only $\|B\|/\lambda_{\min}(H)$, and this can be arbitrarily conservative. Moreover, pointwise first-order data alone are insufficient for exact modulus computation. Thus the exact computation of the Lipschitz modulus without difficult second-order variational information remains open beyond special structures where the inverse sensitivity operator, or an equivalent sharp graph formula, can be computed.

References

- [1] Y. Cui, T. Hoheisel, T. T. A. Nghia, and D. Sun. Lipschitz stability of least-squares problems regularized by functions with C^2 -cone reducible conjugates. *Mathematics of Operations Research*, 2026. arXiv:2409.13118.
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